

Deep Neural Networks: Week 4 Overview

Building Deep Models from Fundamental Components

Syed Hamed Raza

University of Management and Technology (UMT), Lahore

November 22, 2025

Outline

- 1 Introduction
- 2 Shallow vs Deep Models
- 3 Deep Network Structure
- 4 Notation
- 5 Conclusion

- Transition from logistic regression and shallow networks.
- Review of forward and backpropagation fundamentals.
- Motivation for deeper architectures.
- Connection with real-world high-dimensional learning tasks.

Shallow vs Deep Neural Networks

- Logistic Regression = 1-layer neural network.
- A single hidden layer = 2-layer neural network.
- Deep networks: multiple hidden layers capturing hierarchical features.
- Depth as a hyperparameter: evaluated via dev/test performance.

What is a Deep Neural Network?

- A model with more than one hidden layer.
- Enables hierarchical feature extraction.
- Can represent complex nonlinear relationships.
- Empirically successful in vision, NLP, speech, and multimodal tasks.

Layer Notation

- L : total number of layers (excluding input layer).
- $n^{[l]}$: number of units in layer l .
- $a^{[l]}$: activations of layer l , with $a^{[0]} = x$.
- $W^{[l]}, b^{[l]}$: weights and biases for layer l .
- $a^{[L]} = \hat{y}$: network prediction.

Example Network Structure

- Input: $n^{[0]} = 3$ features.
- Hidden Layer 1: $n^{[1]} = 5$ units.
- Hidden Layer 2: $n^{[2]} = 5$ units.
- Hidden Layer 3: $n^{[3]} = 3$ units.
- Output Layer: $n^{[4]} = 1$ unit.

Key Takeaways

- Deep networks extend shallow networks by stacking multiple layers.
- Notation $(W^{[l]}, b^{[l]}, a^{[l]})$ is essential for implementation.
- Depth enables hierarchical learning but increases complexity.
- Forward propagation generalizes naturally to multiple layers.